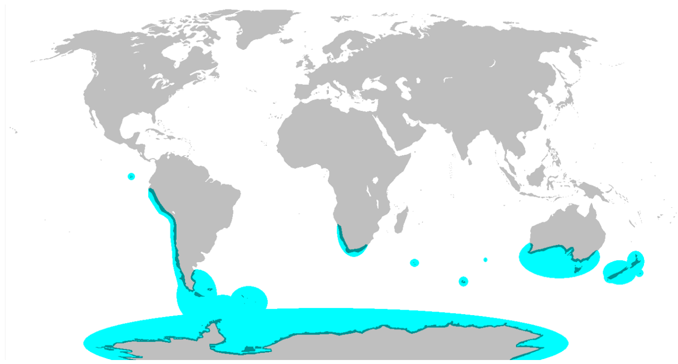
Study Guide One
Topics: Biodiversity, Climate and Biomes, Allocation and Tradeoffs, Resource Acquisition, Osmoregulation, Thermoregulation and Niches

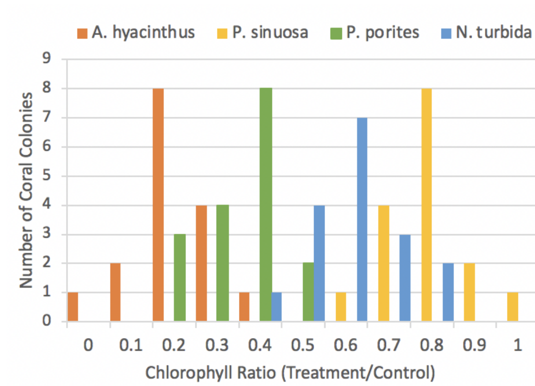
This study guide provides examples for the topics that you might see on the first midterm. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and to review all the topics that have been covered.

1. The map below shows the distribution of penguin species on Earth in blue (landmasses are shown in gray). This distribution can best be explained by…
    a. Movement of tectonic plates
    b. Dispersal of ancestral penguin species to unconnected landmasses
    c. Independent evolution of penguin species in different regions
    d. The rainshadow effect



2. Consider the number of species within a given taxonomic group, of any level (kingdom, phylum, etc). Select all the statements below that are true.
    a. A class will contain at least as many species as an order within that class contains
    b. A phylum will always contain more species than any class or order
    c. All orders contain the same number of families
    d. A taxonomic grouping of any level could contain only a single species

3. The histogram below shows the chlorophyll ratios (x axis) for four species of coral. The chlorophyll ratio is calculated as the amount of chlorophyll in warming treated corals divided by the amount of chlorophyll in control corals. Chlorophyll is produced by the coral's algal partner and so higher chlorophyll levels indicate more algal partners present in the coral colony. **According to the data shown in the graph, which coral species has the highest heat tolerance?**
    a. A. hyacinthus
    b. P. sinuosa
    c. P. porites
    d. N. turbida
    e. All species shown here have equal heat tolerance



4. You are developing a study on the effect of increasing ocean acidification (pH levels) on the mortality rates of a marine fish species. You expose individuals of the species to two different pH conditions: normal pH levels the fish regularly experience in their habitat and lower pH values representing the potential impacts of acidification. At the end of the treatments, you measure mortality rates in both populations. **For each statement below, determine if the statement is an alternate hypothesis, a null hypothesis, or not an appropriate hypothesis for this study.**
    a. Fish mortality is the same under high and low pH conditions
    This hypothesis is: *NULL* *ALTERNATE* *NOT APPROPRIATE*

    b. Higher pH levels are good for fish mortality rates
    This hypothesis is: *NULL* *ALTERNATE* *NOT APPROPRIATE*

    c. Higher fish mortality increases ocean pH levels
    This hypothesis is: *NULL* *ALTERNATE* *NOT APPROPRIATE*

    d. Higher ocean pH levels increase fish mortality rates
    This hypothesis is: *NULL* *ALTERNATE* *NOT APPROPRIATE*

    e. Higher ocean pH levels decrease fish mortality rates
    This hypothesis is: *NULL* *ALTERNATE* *NOT APPROPRIATE*

5. For each item below, determine whether it is an example of weather or climate
    a. An atmospheric river that drops heavy rainfall in Davis
    This is: *WEATHER* *CLIMATE*

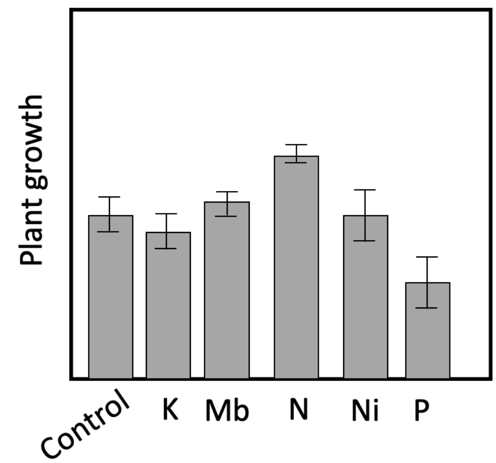
    b. A tornado in Kansas
    This is: *WEATHER* *CLIMATE*

    c. Lower average annual precipitation in deserts than in the tropics
    This is: *WEATHER* *CLIMATE*

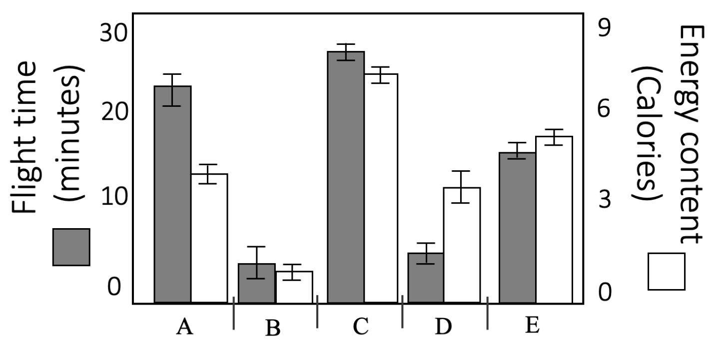
    d. Higher fire risk in California than in Kansas
    This is: *WEATHER* *CLIMATE*

6. How do greenhouse gasses like carbon dioxide affect the earth's radiation budget?
    a. They increase the amount of radiation that is retained in the atmosphere
    b. They increase the amount of radiation that is reflected by clouds into space
    c. They increase the amount of radiation that the earth receives from the sun

7. You are interested in determining the limiting nutrient of plants in a garden. You collect the data in the figure below by comparing growth under control conditions (with no additional nutrients) to growth of plants with added nitrogen, phosphorus, molybdenum, potassium, or nickel. In this ecosystem, which nutrient is limiting?
    a. Phosphorous (P)
    b. Nitrogen (N)
    c. Molybdenum (Mb)
    d. Potassium (K)
    e. Nickel (Ni)
    f. None of these nutrients are limiting in this environment



8. You are studying which meadows a hive of honeybees visits for nectar foraging. You collect the data below for five different meadows (A-E, x axis) including energy investment in accessing the meadow, in terms of honeybee flight time (gray bars, left y axis) and ease of accessing nectar in the flower; as well as the energy content of nectar in flowers blooming in each meadow in Calories (white bars, right y axis).
Which meadow does optimal foraging theory predict the honeybees should visit?
    a. Meadow A
    b. Meadow B
    c. Meadow C
    d. Meadow D
    e. Meadow E
    f. All five meadows are equally beneficial in terms of optimal foraging theory predictions



9. Anadromous fish are unusual in that they are able to migrate between marine and freshwater environments throughout their life, which requires a shift in the osmoregulatory mechanisms by which the fish maintains an appropriate balance of water and salt in its body. In the marine environment, the ocean is _[blank1]_ compared to the fish's body and water will move _[blank2]_.

    Options for blank 1: hypotonic, hypertonic, isotonic
    Options for blank 2: into the fish's body, out of the fish's body

10. Anadromous fish such as salmon spend the majority of their lives in the ocean (which is saltwater), but migrate to rivers (which are freshwater) to mate and reproduce. During this migration, salmon spend a period of time at the mouth of the river to allow their osmoregulatory mechanisms to adjust to the change in salinity between the ocean and the river. This is an example of…
    a. Acclimation
    b. Adaptation

11. Wild turkeys are commonly found around Yolo County, including in the city of Davis. While the adult birds can tolerate a wide temperature and precipitation range, juvenile turkeys (eggs and hatchlings) are more susceptible to cold, especially under wet conditions. Wild turkeys eat a wide variety of foods, including seeds and berries, as well as small insects and other invertebrates. These types of food are especially common in agricultural fields in Yolo County, meaning wild turkeys commonly roost in portions of the county with a high prevalence of agricultural lands. Some regions of the county, such as the town of Winters, are home to wild peafowl, which have similar abiotic tolerances and food sources as wild turkey. Though smaller than wild turkeys, peafowl are aggressive enough to outcompete the wild turkey for food and space.
    Based on the description above, which of the following statements are true about the niche of wild turkeys? *Select all that apply*
    a. The fundamental niche of wild turkeys changes throughout their lifespan
    b. The realized niche of wild turkeys excludes regions inhabited by wild peafowl
    c. The fundamental niche of wild turkeys is influenced by human agricultural activities
    d. The realized niche of wild turkeys is reduced compared to its fundamental niche
    e. The realized niche of wild turkeys is influenced by the amount of precipitation in winter

12. Many organisms use evaporation of water as a cooling mechanism. For example, humans produce sweat, dogs pant, and plants transpire water through pores in their leaves. While this strategy is effective for reducing body temperature, what trade-off is it most likely to impose?
    a. A reduction in the nutrients available to invest in defensive compounds for avoiding predation or herbivory
    b. An increase in the energy required to complete other life processes such as reproduction
    c. An increased risk of dehydration or desiccation
    d. A reduction in the fundamental niche of the species to regions with cooler temperatures

13. Scientists discover a new species of fish in a deep ocean trench and biologists studying them collect the following life history data: The fish are very large, nearly 200 tons in weight and over 30 meters in length. Their life span is about 800 years and they take nearly 100 years to reach sexual maturity. After mating, each female lays thousands of eggs in a nest on the ocean floor that she does not revisit or maintain after laying the eggs. Once the eggs hatch, many of the juveniles succumb to disease or predation and do not survive to maturity.
    How would you describe this species in terms of *r* and *K* selection?
    a. These traits are all *r*-selected
    b. These traits are all *K*-selected
    c. These traits are in-between *r* and *K* selection (in the middle of continuum, not at either extreme)
    d. This species has some *r*-selected traits and some *K*-selected traits (at both ends of the continuum, representing both extremes)

14. Put these terms in order from most broad to most specific: phylum, species, population, kingdom, class, family, order, genus, individual

15. Compare and contrast species richness and species evenness.

16. What process drives the difference in vegetation between the western- and eastern-facing slopes of the Sierra Nevada mountains?

17. How does the tilt of the Earth's axis drive seasons?

18. How do current concentrations of greenhouse gases (carbon dioxide, methane, and nitrous oxide) compare to historic levels?

19. Using the Principle of Allocation, explain the concept of a trade-off. Give an example of a tradeoff involving adaptations or strategies for:
    a. Resource acquisition
    b. Reproduction

20. Compare and contrast the following terms:
    a. Evolution and ecology
    b. Species richness, species evenness, and species diversity
    c. Endotherm and ectotherm
    d. Weather and climate
    e. *Rhizobia* and mycorrhizae
    f. Fundamental niche and realized niche
    g. r-selected traits/organisms and K-selected traits/organisms
    h. hypertonic, hypotonic, isotonic

21. Using your understanding of osmoregulation, explain why the Hawaiian Islands have no native amphibians.

22. As CO_2 began to rise, there was a hypothesis that plants would photosynthesize more, removing the additional CO_2 from the air. Use your knowledge of plant resource acquisition to propose an explanation for why increased CO_2 in the atmosphere might **not** significantly increase photosynthetic rates in plants.

23. Humans have a fondness for sugary and fatty foods. How does optimal foraging theory explain the evolution of this preference? Given our current ecological conditions, what are the consequences of this preference?

24. Imagine you are doing research on plant populations at Jepson Prairie. How would you determine what resources limit plant growth in this environment?

25. Assign each of the climatographs below to a biome and to a hemisphere (northern, southern, or equatorial). Possible biomes include deserts, grasslands, tropical rainforests, temperate forests, boreal forests, tundra, and Mediterranean climes. Be prepared to explain your decision.